ORDINARY DIFFERENTIAL EQUATIONS • UNIT 1.01

C First-Order Separable Equations & Existence Theory

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

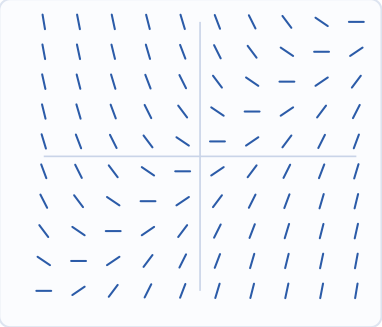
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 D	Q3 D	Q4 A	Q5 A
Q6 A	Q7 D	Q8 A	Q9 C	Q10 D
Q11 A	Q12 B	Q13 B	Q14 D	Q15 A
Q16 D	Q17 C	Q18 C	Q19 A	Q20 C
Q21 D	Q22 C	Q23 A	Q24 B	Q25 A

First-Order Separable Equations & Existence Theory Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



- **A** 39 ✓
- **B** 24
- **C** 42
- **D** 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is 39.

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

B $e^2 - 1$

C $3 - 2e^{-2}$

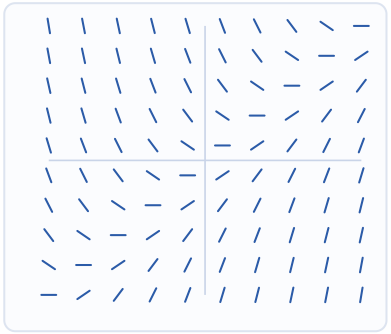
D $3 - 2e^{-1}$ ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$.
Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$.
Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$.
The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$.
Integrating both sides with respect to x: $ye^{2x} = \int 6e^{2x}dx$.
$ye^{2x} = 6 \cdot \frac{1}{2}e^{2x} + C$.
$ye^{2x} = 3e^{2x} + C$.
Solving for y: $y = 3 + Ce^{-2x}$.
Using the initial condition $y(0) = 1$ to find C: $1 = 3 + Ce^{-2(0)}$.
$1 = 3 + C$.
$C = 1 - 3 = -2$.
So, the particular solution is $y(x) = 3 - 2e^{-2x}$.
Now, we need to find $y(1)$: $y(1) = 3 - 2e^{-2(1)} = 3 - 2e^{-2}$.
This matches choice C.

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D **1.220** ✓

WORKED SOLUTION & VERIFICATION

Using Euler's method $y_{n+1} = y_n + h(x_n + y_n)$ with $x_0 = 0, y_0 = 1, h = 0.1$:
1. $y_1 = 1 + 0.1(0 + 1) = 1.100$ at $x_1 = 0.1$.
2. $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.220$ at $x_2 = 0.2$.

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

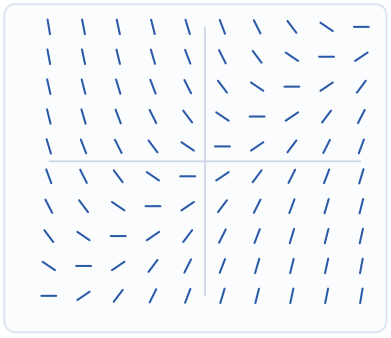
- **A** 5 ✓
- **B** -5
- **C** 6
- **D** -6

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation $r^2 - 5r + 6 = 0$, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $r_1 + r_2 = -\frac{-5}{1} = 5$.

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



- **A** 7 ✓
- **B** 6
- **C** 8
- **D** 5

WORKED SOLUTION & VERIFICATION

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular solution, we have $y(2) = (2)^2 + 3 = 4 + 3 = 7$.

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

- **A** $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓
- **B** $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$
- **C** $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$
- **D** $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of x to be the same. Let $k = n-2$ in the first series, so $n = k+2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n+1$ in the second series, so $n = k-1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all x, the coefficients of each power of x must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing k with n, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** 24
- **D** 40 ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is 40.

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

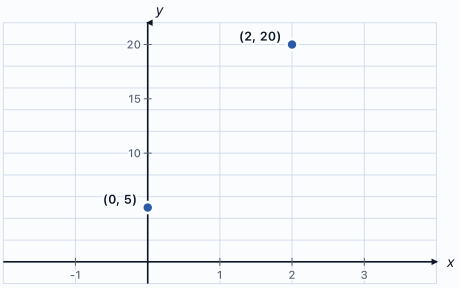
- **A** 170 ✓
- **B** 151
- **C** 173
- **D** 187

WORKED SOLUTION & VERIFICATION

Substituting $s = 20$ gives $\frac{170}{20-19} = 170$.

9 MCQ

Consider the separable differential equation $\frac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.



A $k = \ln 2, y(3) = 40$

B $k = 2\ln 2, y(3) = 40$

C $k = \ln 4, y(3) = 80$ ✓

D $k = \ln 2, y(3) = 20\sqrt{2}$

WORKED SOLUTION & VERIFICATION

The differential equation $\frac{dy}{dx} = ky$ is separable. Separating variables gives $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides gives $|y| = e^{kx+C_1} = e^{C_1}e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. \n\nWe are given that the solution curve passes through $(0, 5)$. Substituting these values into $y = Ce^{kx}$: $5 = Ce^{k(0)} \Rightarrow 5 = C$. Thus, the particular solution is $y = 5e^{kx}$. \n\nWe are also given that the solution curve passes through $(2, 20)$. Substituting these values into $y = 5e^{kx}$: $20 = 5e^{k(2)} \Rightarrow 4 = e^{2k}$. Taking the natural logarithm of both sides: $\ln 4 = \ln(e^{2k}) \Rightarrow \ln 4 = 2k \Rightarrow k = \frac{\ln 4}{2} = \frac{2\ln 2}{2} = \ln 2$. \n\nSo, the value of k is $\ln 2$. The particular solution is $y = 5e^{(\ln 2)x} = 5(e^{\ln 2})^x = 5(2^x)$. \n\nFinally, we need to find the value of y when $x = 3$. Substituting $x = 3$ into $y = 5(2^x)$: $y(3) = 5(2^3) = 5(8) = 40$. \n\nTherefore, $k = \ln 2$ and $y(3) = 40$.

10 MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 45

B 50

C 23

D **42** ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 42\lambda + 272 = 0$, the sum of roots is 42.

11 MCQ

Consider the separable differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with the initial condition $y(1) = 2$. Find the value of $y(3)$.

A $2\sqrt{5}$ ✓

B $2\sqrt{3}$

C $2\sqrt{2}$

D 4

WORKED SOLUTION & VERIFICATION

The differential equation is separable: $ydy = 2xdx$.

Integrate both sides: $\int ydy = \int 2xdx$.

This gives $\frac{1}{2}y^2 = x^2 + C$.

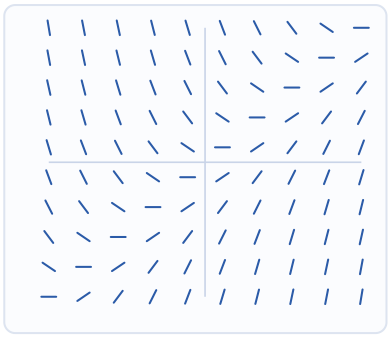
Use the initial condition $y(1) = 2$ to find C: $\frac{1}{2}(2)^2 = (1)^2 + C \implies 2 = 1 + C \implies C = 1$.

So, the solution is $\frac{1}{2}y^2 = x^2 + 1$, which means $y^2 = 2x^2 + 2$, and $y = \sqrt{2x^2 + 2}$ (since $y(1) = 2 > 0$).

Now, find $y(3)$: $y(3) = \sqrt{2(3)^2 + 2} = \sqrt{2(9) + 2} = \sqrt{18 + 2} = \sqrt{20} = 2\sqrt{5}$.

12 MCQ

Consider the differential equation $\frac{dy}{dx} = kx(y^2 - 1)$, where k is a positive constant. If the solution passing through $(1, 2)$ also passes through $(2, 3)$, determine the value of the constant k.



A $\frac{1}{3}\ln(\frac{3}{2})$

B $\frac{1}{4}\ln(\frac{2}{3})$ ✓

C $\frac{1}{4}\ln(\frac{3}{2})$

D $\frac{1}{3}\ln(\frac{2}{3})$

WORKED SOLUTION & VERIFICATION

The given differential equation is separable: $\frac{dy}{y^2-1} = kx\,dx$.

Integrating both sides:

$\int \frac{dy}{y^2-1} = \int kx\,dx$

Using partial fractions for the left side, $\frac{1}{y^2-1} = \frac{1}{2}(\frac{1}{y-1} - \frac{1}{y+1})$.

So, $\frac{1}{2}\int(\frac{1}{y-1} - \frac{1}{y+1})\,dy = \frac{1}{2}\ln\left|\frac{y-1}{y+1}\right|$.

The right side integrates to $\frac{1}{2}kx^2 + C_1$.

Thus, $\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = \frac{1}{2}kx^2 + C_1$, which simplifies to $\ln\left|\frac{y-1}{y+1}\right| = kx^2 + C$, where $C = 2C_1$.

Exponentiating both sides gives $\left|\frac{y-1}{y+1}\right| = e^{kx^2+C} = e^C e^{kx^2}$. Let $A = \pm e^C$, then $\frac{y-1}{y+1} = Ae^{kx^2}$.

We are given that the solution passes through $(1, 2)$. Substitute these values:

$\frac{2-1}{2+1} = Ae^{k(1)^2}$

$\frac{1}{3} = Ae^k$

So, $A = \frac{1}{3}e^{-k}$.

Substitute A back into the general solution:

$\frac{y-1}{y+1} = \frac{1}{3}e^{-k}e^{kx^2} = \frac{1}{3}e^{k(x^2-1)}$.

We are also given that the solution passes through $(2, 3)$. Substitute these values:

$\frac{3-1}{3+1} = \frac{1}{3}e^{k(2^2-1)}$

$\frac{2}{4} = \frac{1}{3}e^{k(3)}$

$\frac{1}{2} = \frac{1}{3}e^{3k}$

Multiply by 3: $\frac{3}{2} = e^{3k}$.

Take the natural logarithm of both sides:

$\ln(\frac{3}{2}) = 3k$

Solve for k: $k = \frac{1}{3}\ln(\frac{3}{2})$.

13 MCQ

The separable initial value problem $\frac{dh}{du} = 30u$ satisfies $h(0) = 3$. Determine the solution value $h(3)$.

- A 141
- **B 138** ✓
- C 153
- D 135

WORKED SOLUTION & VERIFICATION

Integrating gives $h(u) = 15u^2 + 3$, so $h(3) = 138$.

14 MCQ

The first-order differential equation $\frac{dh}{du} = 23u - 2h$ has its slope field plotted below. Determine the slope $\frac{dh}{du}$ at the point $X(14, 4)$.

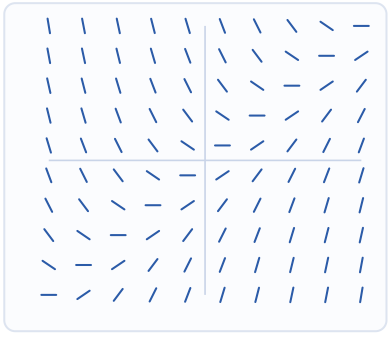
- A 317
- B 310
- C 337
- **D 314** ✓

WORKED SOLUTION & VERIFICATION

Evaluating at $(14, 4)$ gives $23(14) - 2(4) = 314$.

15 MCQ

Consider the separable first-order differential equation $\frac{dy}{dx} = \frac{x^2}{2y}$. If the solution passes through the point $(2, \sqrt{5})$, find the explicit solution for $y(x)$ assuming $y > 0$.



- **A** $y = \sqrt{\frac{x^3}{3} + \frac{7}{3}}$ ✓
- B $y = \sqrt{\frac{x^3}{3} + 2}$
- C $y = \sqrt{\frac{x^3}{3} + \frac{2}{3}}$
- D $y = \sqrt{\frac{x^3}{3} + \frac{5}{3}}$

WORKED SOLUTION & VERIFICATION

Separating variables gives $2y\,dy = x^2\,dx$. Integrating both sides yields:

$$y^2 = \frac{x^3}{3} + C.$$

Substituting $(2, \sqrt{5})$ gives $5 = \frac{8}{3} + C \implies C = \frac{7}{3}$. Since $y > 0$, $y = \sqrt{\frac{x^3}{3} + \frac{7}{3}}$.

16 MCQ

The separable initial value problem $\frac{dU}{dv} = 28v$ satisfies $U(0) = 9$. Determine the solution value $U(16)$.

A 3584

B 3607

C 3596

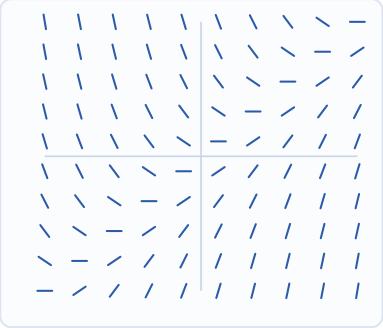
D **3593** ✓

WORKED SOLUTION & VERIFICATION

Integrating gives $U(v) = 14v^2 + 9$, so $U(16) = 3593$.

17 MCQ

The separable initial value problem $\frac{dP}{dr} = 34r$ satisfies $P(0) = 8$. Determine the solution value $P(7)$.



A 858

B 844

C **841** ✓

D 833

WORKED SOLUTION & VERIFICATION

Integrating gives $P(r) = 17r^2 + 8$, so $P(7) = 841$.

18 MCQ

The first-order differential equation $\frac{dR}{dm} = 9m - 5R$ has its slope field plotted below. Compute the slope $\frac{dR}{dm}$ at the point $M(15, 24)$.

A 18

B -9

C **15** ✓

D 24

WORKED SOLUTION & VERIFICATION

Evaluating at $(15, 24)$ gives $9(15) - 5(24) = 15$.

19 MCQ

The first-order differential equation $\frac{dh}{du} = 21u - 3h$ has its slope field plotted below. Compute the slope $\frac{dh}{du}$ at the point $X(18, 22)$.

- **A** **312** ✓
- **B** 290
- **C** 333
- **D** 315

WORKED SOLUTION & VERIFICATION
Evaluating at $(18, 22)$ gives $21(18) - 3(22) = 312$.

20 MCQ

The first-order differential equation $\frac{du}{ds} = 17s - 7u$ has its slope field plotted below. Compute the slope $\frac{du}{ds}$ at the point $K(2, 24)$.

- **A** -158
- **B** -117
- **C** **-134** ✓
- **D** -131

WORKED SOLUTION & VERIFICATION
Evaluating at $(2, 24)$ gives $17(2) - 7(24) = -134$.

21 MCQ

The first-order differential equation $\frac{dF}{dw} = 12w - 4F$ has its slope field plotted below. Compute the slope $\frac{dF}{dw}$ at the point $D(6, 4)$.

- **A** 52
- **B** 59
- **C** 68
- **D** **56** ✓

WORKED SOLUTION & VERIFICATION
Evaluating at $(6, 4)$ gives $12(6) - 4(4) = 56$.

22 MCQ

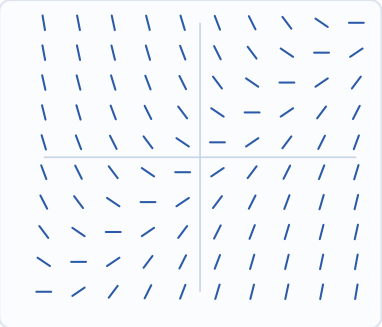
The first-order differential equation $\frac{dM}{dz} = 11z - 1M$ has its slope field plotted below. Compute the slope $\frac{dM}{dz}$ at the point $G(13, 26)$.

- **A** 120
- **B** 91
- **C** **117** ✓
- **D** 128

WORKED SOLUTION & VERIFICATION
Evaluating at $(13, 26)$ gives $11(13) - 1(26) = 117$.

23 MCQ

The first-order differential equation $\frac{dS}{db} = 4b - 5S$ has its slope field plotted below. Determine the slope $\frac{dS}{db}$ at the point $W(7, 17)$.



- **A** -57 ✓
- **B** -53
- **C** -54
- **D** -74

WORKED SOLUTION & VERIFICATION
Evaluating at $(7, 17)$ gives $4(7) - 5(17) = -57$.

24 MCQ

The first-order differential equation $\frac{dp}{dt} = 24t - 6p$ has its slope field plotted below. Compute the slope $\frac{dp}{dt}$ at the point $P(17, 25)$.

- **A** 233
- **B** 258 ✓
- **C** 261
- **D** 282

WORKED SOLUTION & VERIFICATION
Evaluating at $(17, 25)$ gives $24(17) - 6(25) = 258$.

25 MCQ

The first-order differential equation $\frac{df}{dx} = 4x - 2f$ has its slope field plotted below. Compute the slope $\frac{df}{dx}$ at the point $A(10, 21)$.

- **A** -2 ✓
- **B** 2
- **C** 1
- **D** -23

WORKED SOLUTION & VERIFICATION
Evaluating at $(10, 21)$ gives $4(10) - 2(21) = -2$.